

Grant Fonseca

Software Engineer

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SUMMARY

Full-stack software engineer with 6+ years building scalable web applications, REST APIs, event-driven pipelines, and data-driven systems. Track record of owning production features end-to-end — data modeling, API design, automated detection, and alerting. Experienced shipping high-impact systems in fast-moving cross-functional teams with strong emphasis on observability, error handling, and maintainability. Active practitioner of LLMs, agentic workflows, Model Context Protocol (MCP) tool servers, and AI-powered automation as a force multiplier for engineering velocity.

TECHNICAL SKILLS

Languages:	Python, TypeScript, JavaScript, C#, Java, SQL, HTML/CSS
Frontend:	React.js, React Native, Node.js, Express.js
Backend & APIs:	REST API design, MySQL, MongoDB, SQLite, Spring, SQL triggers & stored procedures
Infrastructure:	AWS, CI/CD pipelines, GitHub, Docker, IIS, Linux
Data & Analytics:	Splunk, Apache Kafka, Redis, FME, behavioral analytics, dashboard development
Algorithms & Geo:	Graph theory (Union-Find, Dijkstra), TSP heuristics (nearest-neighbor, 2-opt), OSMnx, NetworkX, geopy
AI & LLMs:	GitHub Copilot, ChatGPT, Claude, Deepseek, Ollama, Model Context Protocol (MCP) server development, Prompt Engineering, LLM Integration, Agentic Workflows, AI Orchestration, NLP, Generative AI, AI-Powered Automation
Tools:	Neovim/Vim, IntelliJ IDEA, PyCharm, Visual Studio, Android Studio
Methodologies:	Agile, DevSecOps, unit & integration testing, change control, event-driven architecture, real-time detection, behavioral analysis, end-to-end ownership

EXPERIENCE

Department Programmer Analyst · City of Santa Rosa Oct 2023 – Present

- Engineered a Model Context Protocol (MCP) server exposing custom AI-agent tools — automated email drafting/sending and live weather data retrieval — over Streamable HTTP, letting the department's FME ETL platform invoke LLM-driven automation as native pipeline steps.
- Designed an intelligent multi-attendee meeting scheduler that infers each attendee's working hours from calendar history via Outlook COM automation, cross-checks shared calendars for conflicts, and books the first mutually available slot — fully non-blocking via multithreaded background execution.
- Built a Union-Find (disjoint-set, union-by-rank with path compression) graph engine and REST API that ranks utility assets by direct/indirect downstream dependency count, surfacing the highest-impact assets before field crews perform replacements.
- Engineered a geospatial route-optimization API combining Dijkstra shortest-path search with TSP heuristics (nearest-neighbor construction plus 2-opt refinement) over OSMnx/NetworkX road-network graphs, cutting multi-stop work-order travel distance and generating one-click optimized Google Maps routes for field technicians.
- Designed and deployed a Python/FME enterprise integration for bi-directional data transfer between asset management platforms via REST APIs — built with composable, higher-order pipeline logic, SQLite-based transaction parsing, and subprocess-isolated execution for scalability and maintainability.
- Built a Python alerting framework that auto-detects integration failures, classifies error types, and pinpoints the exact failure location — delivering structured email alerts that dramatically cut mean time to resolution.
- Automated migration of thousands of critical records into production with zero data loss, using custom Python validation logic to enforce data integrity end-to-end.
- Built a full-stack C# web application with a RESTful API, SQL-trigger-backed business rule enforcement, and custom access control for bulk ArcGIS/SDE geodatabase inserts — eliminating hours of manual data entry.
- Established engineering best practices across all projects: GitHub feature branching, formal release processes, and comprehensive error logging for full audit trails.

- Mentored a junior developer on system architecture and API design while deploying and maintaining CI/CD pipelines for rapid, stable production delivery.

Full Stack Developer · Office Work Software (OrgChart Now)

Nov 2021 – May 2023

- Developed a JavaScript activity-logging system for a SaaS platform with \$10M+ ARR — enabling behavioral analysis, abuse detection, and data-driven product decisions.
- Built Splunk dashboards visualizing application usage and KPIs for stakeholders, establishing monitoring as a core asset for prioritization.
- Owned the full fix lifecycle for critical production issues affecting top-paying customers — diagnosis, resolution, test criteria, and regression testing with minimal customer disruption.
- Developed React and React Native component library for a multi-tenant SaaS platform, including avatars, cards, and navigation components.
- Authored unit and integration tests across a large codebase to ensure reliability and prevent future regressions.
- Designed a cross-platform HTML/CSS/Java RESTful web app for employee workflow management with full CRUD operations and search functionality.
- Led legacy-to-modern migration via API integrations, redesigning the UX and upgrading all application components.
- Defined MVP scope and architecture; delivered a working end-to-end app aggregating background checks and public records while maintaining data integrity through trusted sources.
- Translated stakeholder requirements into working deliverables, consistently maintaining schedule integrity and meeting milestones.

PORTFOLIO

Software Engineering Portfolio · grantfonseca.com

- Portfolio site built in TypeScript and React — highlighting front-end architecture, component design, and type-safe development practices.

EDUCATION

B.S. in Computer Science · Sonoma State University

LinkedIn Learning: React, Agile Development